9 The Wrist and Hand

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Abstract

MRI of the hand and wrist is particularly challenging because of the size and complexity of the anatomic structures. However, technologic advances have greatly improved image quality, and with appropriate sequences and protocols, MRI can be used to visualize many of the clinically important structures in the hand and wrist with good anatomic detail. 3-T imaging has improved detection of small anatomic structures even further, but most hand and wrist examinations are still performed with 1.5-T magnets, which are adequate in most cases. MRI now has an essential role in evaluating the full spectrum of traumatic, degenerative, and other disorders of the hand and wrist. MRI is the preferred modality for many clinicians when evaluating patients with suspected scaphoid fractures and negative radiographs. MR arthrography can be used to assess the triangular fibrocartilage complex and scapholunate and lunotriquetral ligaments, all of which have been difficult to visualize with conventional imaging. MRI can identify a variety of tendon abnormalities, including acute injuries and degenerative and inflammatory conditions. It is useful in early diagnosis of both Kienböck disease and rheumatoid arthritis and may be of prognostic value in these contexts. MRI is also useful in localizing and staging malignant and infectious conditions of the hand and wrist.

Keywords: carpal tunnel, compression neuropathies, fractures, hand, ligaments, osteonecrosis, rheumatoid arthritis, tendons, triangular fibrocartilage complex, wrist

9.1 Normal Anatomy

9.1.1 Sagittal Anatomy

The wrist is made of the distal radius and ulna, proximal and distal carpal rows, and metacarpal bases. The many osseous articulations allow tremendous range of motion. Injuries to the wrist ligaments can manifest in subtle changes in overall osseous alignment. Sagittal alignment can be assessed on magnetic resonance imaging (MRI) in a number of ways. The central column of the wrist comprises the lunate and capitate. Normally, on sagittal images, a single line can be drawn through the distal radius, central column, and third metacarpal. The capitolunate, radiolunate, and scapholunate angles can also be measured and compared with standard values. Deviations in these measurements from their norms suggest carpal instability.

A number of muscles and tendons are responsible for the full range of wrist and hand motion. Sagittal images are particularly useful for evaluating the digital flexor and extensor tendons, both of which are low-signal structures on T1- and T2-weighted images. The flexor digitorum profundus, flexor digitorum superficialis, and flexor pollicis longus tendons enter and then exit the carpal tunnel as they pass over the carpal bones. Proximally, the flexor digitorum superficialis lies superficial to the flexor digitorum profundus. At the level of the proximal phalanx, the flexor digitorum superficialis splits into two slips at the Camper chiasm, allowing the flexor digitorum profundus to pass more superficially. The two flexor digitorum superficialis slips reapproximate deep to the flexor digitorum profundus and insert individually onto the middle phalanx, and the flexor digitorum profundus inserts onto the distal phalanx. From the metacarpophalangeal joints to the distal phalanges, the digital flexor tendons are contained within the flexor tendon sheaths, which consist of a synovial lining overlain by a pulley system created by segmental bands of fibrous tissue. The five annular and three cruciate pulleys prevent tendon bowstringing and provide a biomechanical advantage during finger flexion. The pulleys are small, and advances in imaging have led to improvements in their imagery.[1] They are best seen on axial sections, in which the thicker annular pulleys are easier to see than the thinner cruciate pulleys.

The extensor tendons pass under the extensor retinaculum and over the dorsum of the distal radius and ulna within six fibroosseous extensor compartments, which are described in detail in “Axial Anatomy.” The anatomy of the digital extensor tendons is unique and can be appreciated on sagittal and axial imagery. Proximal to the proximal interphalangeal joint, each digital extensor tendon divides into one central and two lateral slips. The central slip inserts on the middle phalanx. The two lateral slips become confluent with the lateral bands formed by the intrinsic muscles to produce the two conjoined lateral bands. The conjoined lateral bands then converge to form the terminal extensor tendon, which inserts onto the distal phalanx.

The volar plates are fibrocartilaginous thickenings of the volar joint capsules of the metacarpophalangeal and interphalangeal joints. They are seen on sagittal imaging as low-intensity triangular structures. The sagittal bands originate at the volar plates and then extend dorsally where they stabilize the extensor tendons over the metacarpophalangeal joints. The sagittal bands, like the pulley system, are best seen on axial sections. The collateral ligaments of the digits have attachments to the volar plates and are most apparent on coronal sections. The collateral ligaments, together with the volar plates and dorsal capsule, are the primary static stabilizers of the metacarpophalangeal and interphalangeal joints.

9.1.2 Coronal Anatomy

On coronal sections, the proximal and distal carpal rows are delineated by three arcs (one over the proximal extent of the proximal carpal row, one over the distal extent of the proximal carpal row, and one over the proximal extent of the distal carpal row). These arcs, known as Gilula lines, should be smooth and parallel. Disruption of these parallel arcs indicates carpal instability. A different measure of alignment that can be assessed on coronal sections is ulnar variance. Ulnar variance describes the relative lengths of the ulna and the radius, and although it is usually measured on radiographs, it can also be estimated on coronal sections. A difference in ulnar variance of more than 2 mm in the positive or negative direction is considered abnormal. Positive ulnar variance has been associated with

triangular fibrocartilage complex injuries and ulnar impaction syndrome, and negative ulnar variance has been associated with Kienböck disease.

Most of the major intrinsic and extrinsic ligaments of the wrist can be seen on coronal sections. The most clinically important intrinsic wrist ligaments are the scapholunate and lunotriquetral ligaments. These c-shaped structures vary in thickness throughout their courses. Although carpal ligaments usually have low signal intensity on MRI, the scapholunate ligament tends to be more heterogeneous. The scapholunate and lunotriquetral ligaments are made of volar, dorsal, and membranous bands. The dorsal layer has lower signal intensity than the middle or volar layer. Mechanically, the dorsal layer of the scapholunate ligament and the volar layer of the lunotriquetral ligament are most important for joint stability.

Volar and dorsal extrinsic ligaments can also be observed on coronal images.[2] These include the volar radioscaphocapitate ligament, the volar radiolunate ligament (also called the long radiolunate and radiolunotriquetral ligaments), the dorsal intercarpal ligament, and the dorsal radiotriquetral ligament. The volar radiolunate ligament is the largest ligament in the wrist and originates medial and proximal to the volar radioscaphocapitate ligament on the radial styloid. The dorsal intercarpal and dorsal radiotriquetral ligaments attach to the dorsal tubercle of the triquetrum and diverge from one another in a V-shaped configuration to insert on the distal radius and carpus.

The triangular fibrocartilage complex encompasses the major stabilizing structure of the ulnar aspect of the wrist, spanning the distal radioulnar joint and consisting of the following components[3]:

- Dorsal and volar radioulnar ligaments.
- Fibrocartilaginous articular disk.
- Meniscal homologue.
- Extensor carpi ulnaris subsheath.
- Ulnar collateral ligament.
- Ulnolunate ligament.
- Ulnotriquetral ligament.

The normal triangular fibrocartilage complex is a low-signal-intensity, biconcave disk shaped like a bowtie in the coronal plane. The articular disk is readily apparent on coronal and sagittal images. Advances in MRI techniques have enabled visualization of other individual components of the triangular fibrocartilage complex such as the radioulnar ligaments and the extensor carpi ulnaris's subsheath.

The courses of the main arterial structures of the wrist and hand can be traced on coronal images. The radial and ulnar arteries typically anastomose at the deep and superficial palmar arches, although variations exist. After passing through the snuffbox and entering the hand, the radial artery terminates in the deep palmar arch. The ulnar artery enters the hand via Guyon canals and terminates in the superficial palmar arch. The superficial and deep palmar arches provide the main blood supply to the palm and digits.

9.1.3 Axial Anatomy

The axial sections provide optimal imaging of the carpal tunnel and Guyon canal. The carpal tunnel contains the median nerve, flexor digitorum superficialis, flexor digitorum profundus, and flexor pollicis longus tendons. The flexor pollicis longus tendon lies most radially in the tunnel. The four flexor digitorum superficialis tendons lie ulnar to the flexor pollicis longus tendon and superficial to the four flexor digitorum profundus tendons. The roof of the carpal tunnel is formed by the low-signal-intensity transverse carpal ligament, which is deep to the palmar fascia and attaches proximally to the scaphoid tubercle and pisiform and distally to the trapezium and hook of the hamate. The transverse carpal ligament is split into two layers by the flexor carpi radialis tendon, which passes through it before inserting onto the second metacarpal base. The carpal bones form the floor of the tunnel.

Guyon canal lies ulnar and superficial to the carpal tunnel in a depression between the pisiform and the hook of the hamate. Its floor is formed by the transverse carpal ligament and hypothenar muscles, and its roof is formed by the volar carpal ligament. The ulnar nerve and artery traverse the tunnel; the nerve is situated ulnar to the artery. The nerve gives rise to its terminal sensory and deep motor branches within the canal, and three zones of the canal have been defined. The first zone is located proximal to the bifurcation and contains the ulnar nerve proper. The second and third zones extend from the bifurcation distally. The second zone contains the deep motor branch, and the third zone contains the superficial sensory branch. These zones represent distinct sites of nerve compression with unique clinical manifestations.

On the dorsal side of the wrist, axial sections show the six extensor compartments clearly.

- I: Extensor pollicis brevis, abductor pollicis longus.
- II: Extensor carpi radialis brevis, extensor carpi radialis longus.
- III: Extensor pollicis longus.
- IV: Extensor digitorum communis, extensor indicis.
- V: Extensor digiti minimi.
- VI: Extensor carpi ulnaris.

The extensor tendons pass through these fibroosseous canals as they course distally over the dorsal radius and ulna. The extensor retinaculum, an intermediate-signal structure on T1- and T2-weighted sequences, forms the roof of the compartments, which have fixed relationships to the underlying osseous anatomy. Lister tubercle lies between the second and third compartments, the fifth compartment overlies the distal radioulnar joint, and the sixth compartment lies in a deep groove radial to the ulnar styloid.

9.2 Systematic Approach to Image Review

T1-weighted, T2-weighted, and specialized sequences should be viewed in all three imaging planes. T2-weighted images should be examined first for areas of increased signal intensity, which can then be correlated with T1-weighted images for improved anatomic detail. Structures should be comprehensively evaluated in a stepwise fashion. Bones and joints are evaluated first to detect abnormal areas that could represent occult fractures, osteonecrosis, infection, neoplasm, osteophytes, joint space narrowing, erosions, or chondral defects. Attention must

also be paid to overall osseous alignment to detect signs of carpal instability, which can be subtle. This should include inspection of Gilula lines and measurement of capitolunate, radiolunate, and scapholunate angles. Ulnar variance should also be assessed, especially in patients with suspicion for triangular fibrocartilage complex injuries, ulnar impaction, or Kienböck disease. The synovium and synovial fluid should be examined for presence of effusion, loose bodies, or synovial proliferation. The triangular fibrocartilage complex should be assessed for signs of tearing or degeneration. If the study is an arthrogram, the presence or absence of fluid leakage should be noted. The intrinsic and extrinsic ligaments should be examined to determine whether tears or attritional changes are present. The pulleys, collateral ligaments of the digits, volar plates, and sagittal bands should be evaluated if possible. The flexor and extensor tendons should be examined along their lengths for rupture, adhesions, tenosynovitis, or subluxation. Subluxation of the extensor tendons at the metacarpophalangeal joint suggests injury to the sagittal bands. The carpal tunnel should be inspected for enlargement or enhancement of the median nerve or masses within the tunnel. Similarly, Guyon canal should be assessed for the presence of masses, hook of the hamate fractures, or abnormal ulnar nerve changes. Finally, the remaining soft tissues should be checked for fluid collections, neoplasms, or other abnormalities.

9.3 Six Specialized Pulse Sequences and Imaging Protocols

The routine use of MRI for the distal upper extremity, compared with its use for larger joints, has been hindered by the technical challenges of imaging small anatomic structures. MRI of the wrist and hand typically is performed at a magnetic field strength of 0.2 to 3.0 T, using systems with dedicated surface coils. Recent advances in high-field-strength magnets and extremity surface coil design, however, have improved image quality.[1,4] Prospective studies have shown higher contrast-to-noise ratios between muscle and bone and between bone and cartilage and greater detection of anatomic structures with 3-T than with 1.5-T systems.[5,6] The 3-T systems are now becoming available for clinical use, but most hand and wrist examinations are still performed with 1.5-T magnets.

The hand and wrist are best imaged with the patient prone and the affected extremity placed above the head with the forearm pronated. Typically, the closer the body part being imaged is to the isocenter of the magnet, the better the image quality. Because this position is uncomfortable for many patients, scanning may be performed with the patient supine, the arm positioned at the patient's side, and the forearm in neutral position, with the attendant risk of some degradation of image quality. A small field of view and 1- to 3-mm thick sections are preferable.

Scanning is performed routinely in the axial, coronal, and sagittal planes. Most centers use a standard screening examination, consisting of an array of specified imaging planes and their corresponding pulse sequences. According to the indication, other pulse sequences may be added. Standard pulse sequences include the following:

- T1-weighted.
- T2-weighted fast spin-echo (FSE) with fat suppression.
- Short tau inversion recovery (STIR).
- Gradient-echo.

T1-weighted images provide good anatomic detail and are especially useful for identifying fat (given its high signal intensity on these images) in lipomatous masses or atrophied muscles. Fat-suppressed, T2-weighted sequences are the most sensitive for detecting many pathologic conditions by accentuating the presence of abnormal fluid, edema, or hemorrhage related to traumatic injury or an inflammatory process. Gradient-echo sequences, especially with a 3D, thin-section technique, are useful for evaluating the carpal ligaments and articular cartilage (see Chapter 5, Articular Cartilage).

MR arthrography of the wrist can be performed with single-, double-, or triple-compartment (radiocarpal, intercarpal, or distal radioulnar compartments) injection of contrast material. Approximately 2 to 5 mL of contrast material is administered using sterile technique and fluoroscopic guidance. A single-compartment radiocarpal technique is used most often with normal dissemination of contrast material between the proximal carpal row and the distal radius. Fat-suppressed, T1-weighted images are typically obtained in all three planes, in addition to other sequences. MR arthrography of the wrist has proven useful in evaluating the scapholunate ligament, the lunotriquetral ligament, and the triangular fibrocartilage complex.[7]

Intravenous contrast provides valuable information for evaluating a soft tissue mass or possible synovial process involving the wrist and hand, especially when combined with fat-suppressed T1-weighted images. It is useful for differentiating solid and cystic lesions, showing rim-enhancing soft tissue abscesses, and determining bone vascularity in cases of suspected osteonecrosis. Intravenous contrast may also be used to perform indirect arthrography, in which the contrast may migrate into the joint over time. This effect is most notable in an inflamed joint. The information to be gained from these procedures must always be weighed against the additional cost and risk associated with an invasive procedure.

9.4 Traumatic Conditions

Traumatic conditions affecting the wrist and hand that are commonly evaluated by MRI include occult fractures, tears of the triangular fibrocartilage complex, and injuries to the interosseous carpal ligaments and small joint ligaments and tendons.

9.4.1 Occult Fractures of the Scaphoid and Other Bones

Initial radiographs of patients with scaphoid fractures are often negative because of the scaphoid's unique morphology. Therefore, patients with negative radiographs and tenderness in the anatomic snuffbox or scaphoid tubercle after traumatic injury to an outstretched hand may be immobilized presumptively until follow-up radiographs confirm or exclude the diagnosis to avoid the potential negative consequences of a missed fracture.

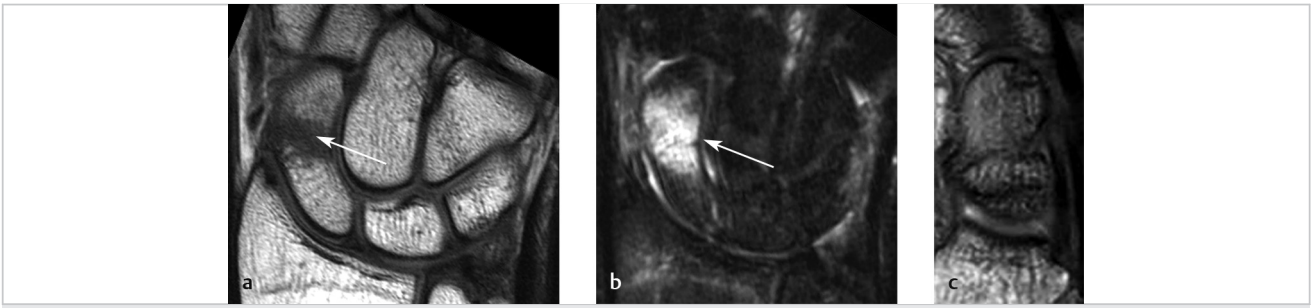


Fig. 9.1 Scaphoid fracture/deformity. **(a)** A coronal T1-weighted image of the wrist showing a fracture through the scaphoid waist (*arrow*), seen as a discrete line of low signal intensity. **(b)** A coronal fat-suppressed T2-weighted image showing the same fracture with focal marrow edema in the scaphoid waist consistent with fracture (*arrow*). **(c)** A sagittal gradient-echo image showing a humpback deformity of the scaphoid. (Reproduced with permission from Brunton LM, Anderson MW, Chhabra AB. The wrist and hand. In: Khanna AJ, ed. MRI for orthopaedic surgeons. New York: Thieme; 2010:129–143.)

Alternatively, advanced imaging modalities can be used to detect radiographically occult scaphoid fracture early to avoid unnecessary immobilization. Early advanced imaging for patients with suspected occult fractures is likely more comfortable and cost-effective than cast immobilization and repeat radiography at a later date.[8] A review of comparative studies sought to determine the superior modality for confirming the diagnosis: bone scintigraphy, CT, MRI, or ultrasonography.[9] Although there is no consensus, many surgeons favor MRI for its safety and ability to provide information about alternative diagnoses.[10,11,12]

A fracture may be identified on T1-weighted images as a distinct line of low signal intensity that contrasts with the surrounding high-signal marrow (▶ Fig. 9.1a). Fat-suppressed, T2-weighted images typically show high-signal marrow edema surrounding the fracture line (▶ Fig. 9.1b). A scaphoid humpback deformity may also be shown on sagittal images (▶ Fig. 9.1c).

MRI is sensitive in the detection of occult fractures in the distal radius, distal ulna, and other carpal bones (▶ Fig. 9.2) and can also differentiate fractures from bone bruises (contusions), which show marrow edema without a discrete fracture line.[13,14] MRI is also sensitive for diagnosing subtle physeal injuries in children (▶ Fig. 9.3) and for evaluating physeal bars and growth arrest.[15] Radiographically, occult fractures of the metacarpals and phalanges are uncommon.



Fig. 9.2 A coronal T1-weighted image of the wrist showing a radiographically occult capitate fracture (*arrows*). (Reproduced with permission from Brunton LM, Anderson MW, Chhabra AB. The wrist and hand. In: Khanna AJ, ed. MRI for orthopaedic surgeons. New York: Thieme; 2010:129–143.)

9.4.2 Triangular Fibrocartilage Complex Injury

Tears of the triangular fibrocartilage complex may be classified as degenerative or traumatic.[16] Degenerative central perforations of the triangular fibrocartilage complex are common and often asymptomatic. Conversely, acute injuries to the triangular fibrocartilage complex frequently lead to ulnar-side wrist pain and dysfunction. It is difficult to distinguish this condition from other causes of ulnar-side wrist pain (such as occult fractures, lunotriquetral ligament injury, extensor carpi ulnaris tendinopathy or subluxation, distal radioulnar joint injury, or pisotriquetral joint injury) by history and physical examination. Historically, three-compartment wrist arthrography was the imaging test of choice for evaluation of the triangular fibrocartilage complex. However, this modality has unacceptably low sensitivity and specificity and lacks the ability to determine lesion size, location, and acuity.[17,18] Although wrist arthroscopy remains the standard for diagnosing triangular fibrocartilage complex pathology, technological advances have improved the ability to evaluate the size, location, and character of triangular fibrocartilage complex tears using MRI and MR arthrography before wrist arthroscopy. In the most specialized centers, the sensitivity and accuracy of MRI for detecting triangular fibrocartilage complex tears approaches 100%, using arthroscopy as the standard diagnostic modality.[19] Not all triangular fibrocartilage complex tears are symptomatic, hence making clinical correlation important. In one study, the prevalence of triangular fibrocartilage complex abnormalities on MRI in 103 asymptomatic volunteers was found to be 38%, with 22% of subjects having complete tears.[20] T2-weighted



Fig. 9.3 Stress injury in the skeletally immature wrist. A sagittal short tau inversion recovery (STIR) image of the wrist showing diffuse marrow edema in the distal radius metaphysis adjacent to the growth plate, which is widened along its volar aspect (*arrow*). (Reproduced with permission from Brunton LM, Anderson MW, Chhabra AB. The wrist and hand. In: Khanna AJ, ed. MRI for orthopaedic surgeons. New York: Thieme; 2010:129–143.)

images may help to differentiate between degenerative and traumatic tears of the triangular fibrocartilage complex, with traumatic tears showing increased signal intensity in adjacent tissues. The excellent anatomic detail afforded by MRI also helps in determining the specific lesion location, with degenerative tears typically occurring in the thinner central portion and traumatic tears most often found in the periphery of the triangular fibrocartilage complex (▶ Fig. 9.4). Nonetheless, all triangular fibrocartilage complex attachments must be carefully evaluated in all three planes. Ulnar attachment tears have been consistently more difficult to diagnose.[21] Concomitant pathology of the extensor carpi ulnaris tendon, lunotriquetral ligament, and surrounding osseous structures must be pursued. MR arthrography may be superior to MRI in detecting some triangular fibrocartilage complex lesions.[22] A triangular fibrocartilage complex tear is often shown by the presence of gadolinium contrast within the tear itself and/or in the distal radioulnar joint. Strong evidence exists that observer experience is critical for the accurate detection of triangular fibrocartilage complex pathology.[23]

9.4.3 Interosseous Ligament Injury

The scapholunate and lunotriquetral ligaments are the most commonly injured wrist ligaments secondary to acute trauma. MRI has supplanted stress radiography and conventional wrist arthrography as the study of choice for suspected carpal ligament injury. Compared with arthroscopy, conventional MRI is highly specific but has an overall low sensitivity for the detection of tears of the scapholunate and lunotriquetral ligaments, although the sensitivity seems to be improving with technologic advances in imaging techniques.[24] Disruption of the interosseous ligaments is best evaluated on coronal gradient-echo (▶ Fig. 9.5) and fat-suppressed T2-weighted images. The normally heterogenous appearance of the scapholunate ligament on coronal images may lead to the misdiagnosis of a tear. Distinct linear areas of high signal intensity equal to that of fluid, especially in the thick dorsal portion of the ligament, usually indicate clinically meaningful tears, which may lead to carpal instability if not treated. Alternative findings in ligament tears include nonvisualization of the ligament or morphologic distortion.[25] MR arthrography may be more accurate than conventional MRI for the detection of interosseous ligament injuries.[4,26,27,28]

9.4.4 Extrinsic Carpal Ligament Injury

The extrinsic carpal ligaments are a complex assortment of dorsal and volar structures that lie in close relation to the wrist joint capsule.[29] The volar extrinsic ligaments are stronger and more robust than their dorsal counterparts. Isolated acute injuries to the extrinsic ligaments are rare. Typically, disruptions of these ligaments are part of a constellation of acute bone and soft tissue injuries about the wrist joint. Although MRI can distinguish the extrinsic ligament complexes with multiple image planes (▶ Fig. 9.6), the role of MRI in this clinical setting is more often to exclude occult fracture, disruption of the intrinsic carpal ligaments, or injury to the triangular fibrocartilage complex. If diagnosed, these conditions may require a change in treatment.

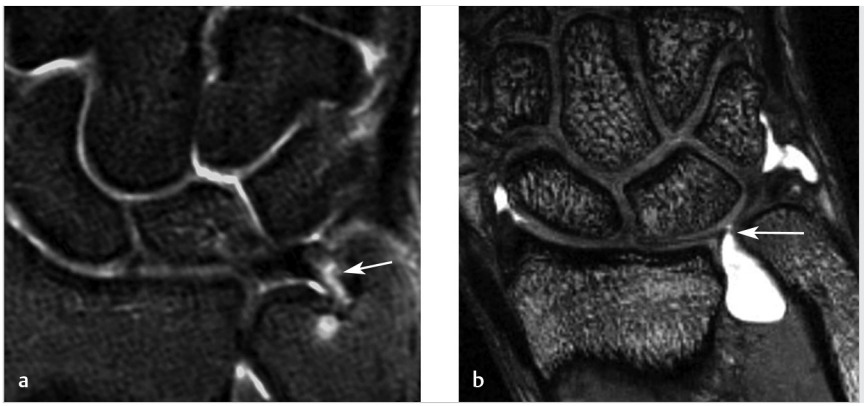


Fig. 9.4 Peripheral triangular fibrocartilage complex tear. **(a)** A coronal fat-suppressed T2-weighted image showing a tear of the peripheral portion of the triangular fibrocartilage complex near its ulnar attachment (*arrow*). **(b)** A coronal gradient-echo image showing a perforation of the central portion of the triangular fibrocartilage complex near its radial attachment (*arrow*). (Reproduced with permission from Brunton LM, Anderson MW, Chhabra AB. The wrist and hand. In: Khanna AJ, ed. MRI for orthopaedic surgeons. New York: Thieme; 2010:129–143.)

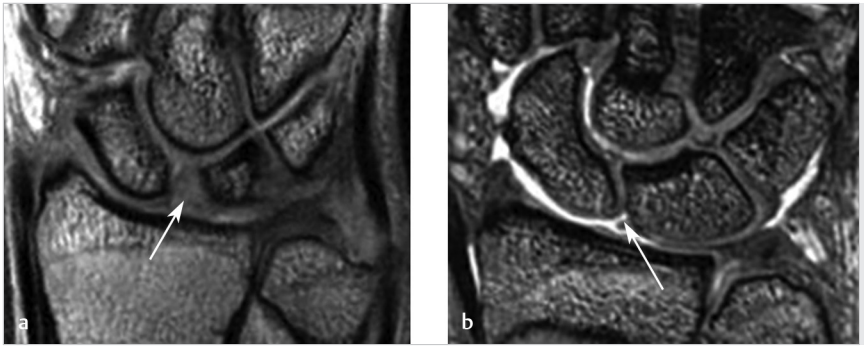


Fig. 9.5 Scapholunate ligament tear. **(a)** A coronal gradient-echo image of the wrist showing partial stretching of the scapholunate ligament with concurrent widening of the scapholunate interval (*arrow*). **(b)** A coronal gradient-echo image of the wrist showing a small perforation in the membranous portion of the scapholunate ligament (*arrow*). (Reproduced with permission from Brunton LM, Anderson MW, Chhabra AB. The wrist and hand. In: Khanna AJ, ed. MRI for orthopaedic surgeons. New York: Thieme; 2010:129–143.)

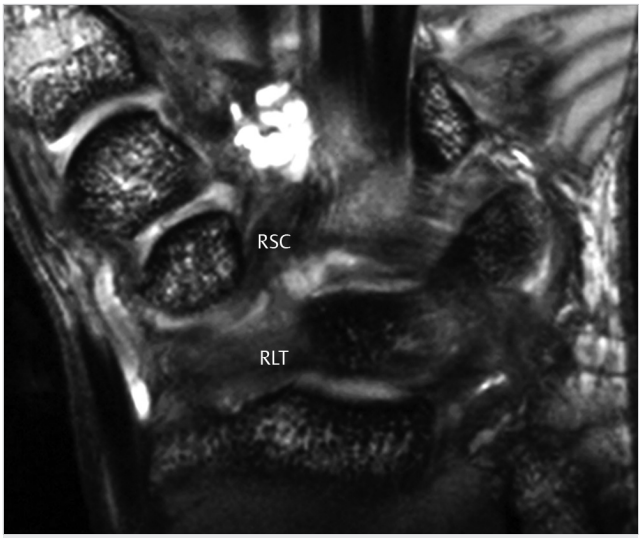


Fig. 9.6 A coronal gradient-echo image of the wrist showing the extrinsic ligaments comprising part of the volar wrist capsule, including the radioscaphocapitate (*RSC*) and radiolunotriquetral (*RLT*) ligaments. (Reproduced with permission from Brunton LM, Anderson MW, Chhabra AB. The wrist and hand. In: Khanna AJ, ed. MRI for orthopaedic surgeons. New York: Thieme; 2010:129–143.)

9.4.5 Thumb Ulnar Collateral Ligament Injury

The ulnar collateral ligament of the metacarpophalangeal joint is the primary stabilizer of the ulnar aspect of the thumb. Rupture of this ligament is common in skiers and results from forced abduction of the thumb. A Stener lesion describes displacement of the torn ligament to a position superficial to the aponeurosis of the adductor pollicis muscle, a conformation that inhibits ligament healing and is an indication for surgery. Although clinical presentation and stress radiography may be enough for the diagnosis of a torn ulnar collateral ligament, they cannot predict the presence of a Stener lesion. MRI helps guide surgical treatment of acute injuries by detecting Stener lesions and allowing measurement of ligament displacement, which has been used in predicting the success of nonoperative treatment (▶ Fig. 9.7).[30,31,32]

9.4.6 Tendon Injuries

Tendon pathology may occur as a result of acute trauma, chronic overuse, or inflammatory arthropathies such as rheumatoid arthritis (RA). The spectrum of pathology ranges from tendon thickening and various degrees of tendon or tendon sheath inflammation to frank tears. In patients with traumatic injury, partial or complete rupture of a hand or wrist tendon may be difficult to diagnose clinically. Although tenderness and localized swelling are often present, the degree of flexion or extension loss may be equivocal. The use of MRI and ultrasonography in this setting is evolving. Occasionally, ruptures of the extensor wrist tendons can be detected by MRI (▶ Fig. 9.8). On T2-weighted sequences, partial tendon tears typically show increased intrasubstance signal intensity (▶ Fig. 9.9), whereas complete tears show separation of the tendon ends with fluid tracking between its margins. Timely and accurate diagnosis has important clinical implications because early surgical repair



Fig. 9.7 A coronal oblique T2-weighted image of the thumb showing a complete rupture of the ulnar collateral ligament at the metacarpophalangeal joint with increased fluid emanating from the joint (*arrow*). (Reproduced with permission from Brunton LM, Anderson MW, Chhabra AB. The wrist and hand. In: Khanna AJ, ed. MRI for orthopaedic surgeons. New York: Thieme; 2010:129–143.)

provides the best outcome for zone-II flexor tendon tears comprising more than 50% of their width.[33] Similarly, identifying the location of the retracted tendon end after a zone-I flexor digitorum profundus avulsion injury (Jersey finger) is important for determining the appropriate treatment. Advances in image quality with sophisticated systems and MR tenography, in which contrast material is injected within the tendon sheath, have led to the detection of flexor pulley system injuries and concomitant bowstringing deformities of the flexor tendons.[1,34]

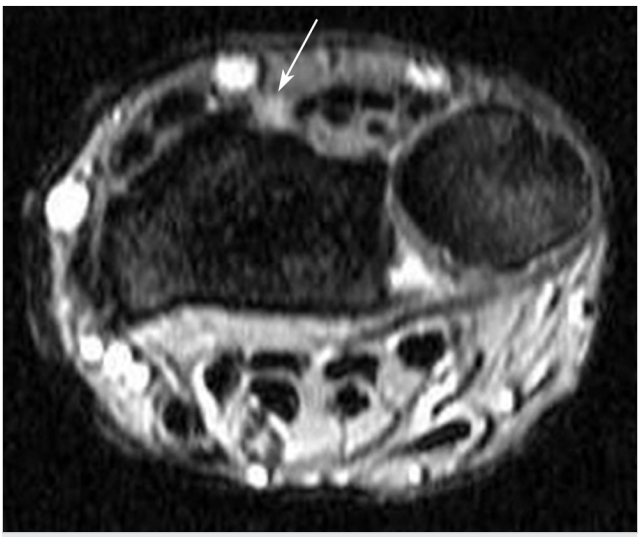


Fig. 9.8 An axial gradient-echo image of the wrist showing absence of the extensor pollicis longus tendon in the third extensor compartment, indicative of complete rupture (*arrow*). (Reproduced from Brunton LM, Anderson MW, Chhabra AB. The wrist and hand. In: Khanna AJ, ed. MRI for orthopaedic surgeons. New York: Thieme; 2010:129–143.)

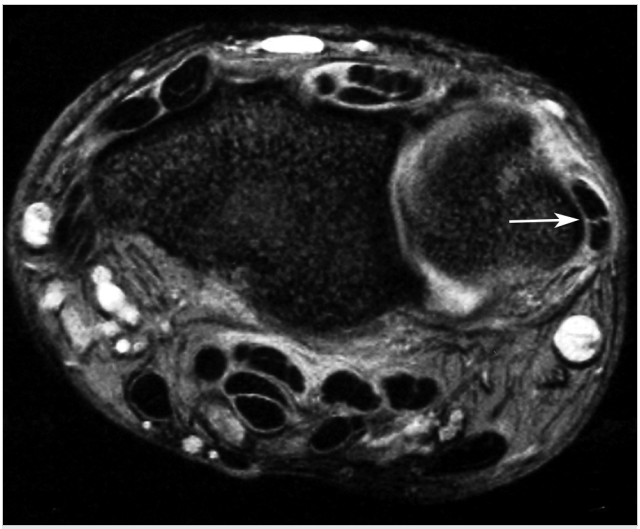


Fig. 9.9 An axial fat-suppressed T2-weighted image at the level of the distal radioulnar joint, showing subluxation of the extensor carpi ulnaris tendon and a high-signal-intensity split within the tendon, indicating a partial tear (*arrow*). (Reproduced with permission from Brunton LM, Anderson MW, Chhabra AB. The wrist and hand. In: Khanna AJ, ed. MRI for orthopaedic surgeons. New York: Thieme; 2010:129–143.)

9.5 Degenerative Conditions

The wrist and hand are affected by various degenerative conditions, although the usefulness of MRI may be limited to evaluating ulnar impaction syndrome, tendon disorders, and carpal instability in select patients. Other common degenerative conditions in the hand and wrist such as first carpometacarpal joint

and radiocarpal joint osteoarthritis are adequately imaged with conventional radiography.

9.5.1 Ulnar Impaction Syndrome

Ulnar variance relates to the relative lengths of the radius and ulna. In patients with ulnar positive variance, the relatively long ulna may abut the proximal carpal row when the wrist is ulnarly deviated, leading to ulnar-side wrist pain from excessive loading across the ulnar aspect of the wrist. Chronic impaction may lead to subchondral sclerosis or cyst formation of the proximal and ulnar portions of the lunate, the radial portion of the triquetrum, and/or the distal and radial portions of the ulnar head. Focal defects of the articular cartilage or increased marrow signal on T2-weighted images may be visualized (▶ Fig. 9.10). These findings can be differentiated from Kienböck disease because the signal changes are focal rather than diffuse, and the ulnar variance is usually positive rather than negative or neutral. MRI may also detect accompanying degenerative triangular fibrocartilage complex or lunotriquetral ligament tears.[35]

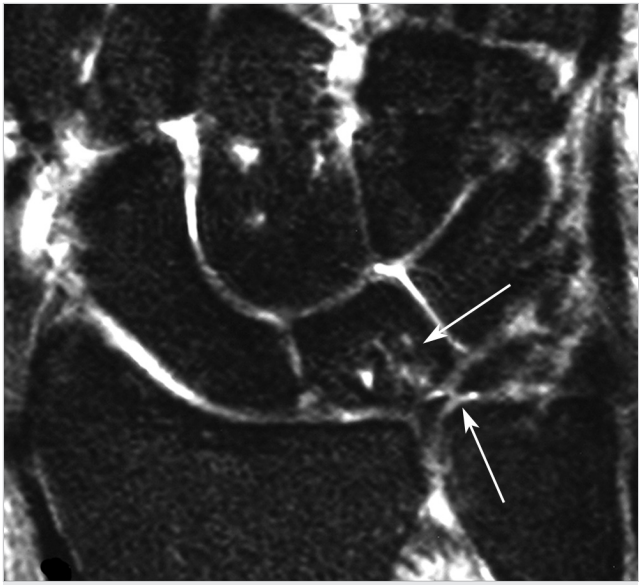


Fig. 9.10 A coronal short tau inversion recovery (STIR) image of the wrist showing focal edema and cystic changes at the proximal ulnar aspect of the lunate consistent with ulnar impaction syndrome (*top arrow*). A small perforation of the central portion of the triangular fibrocartilage complex is also shown (*bottom arrow*). (Reproduced with permission from Brunton LM, Anderson MW, Chhabra AB. The wrist and hand. In: Khanna AJ, ed. MRI for orthopaedic surgeons. New York: Thieme; 2010:129–143.)

9.5.2 Tendon Disorders

Pain and diminished function of the wrist and hand may be caused by various forms of tendon pathology, ranging from tendinosis to tenosynovitis to partial or complete tears. Although these conditions may result from simple overuse, other causes include inflammatory arthropathies, indolent infections, pigmented villonodular synovitis, gout, sarcoidosis, or amyloidosis.[36] Although tendon pathology is often self-limiting, MRI is excellent for assessing the spectrum of tendon pathology. MRI may show tendon thickening or thinning in patients with tendinosis. T2-weighted images may show distended tendon sheaths filled with high-signal-intensity fluid in patients with tenosynovitis (▶ Fig. 9.10). Occasionally, tiny foci of low signal intensity, called *rice bodies*, are seen within the fluid in patients with chronic tenosynovitis.

As an example, de Quervain tenosynovitis is a common inflammatory disorder of the first dorsal compartment. A spectrum of MRI findings may be seen, including thickening of the abductor pollicis longus and extensor pollicis brevis tendons, adjacent fluid collections, and increased intratendinous T2-weighted signal intensity (▶ Fig. 9.11). Similar findings are described for other commonly affected tendons, including the extensor carpi ulnaris, extensor carpi radialis longus, extensor

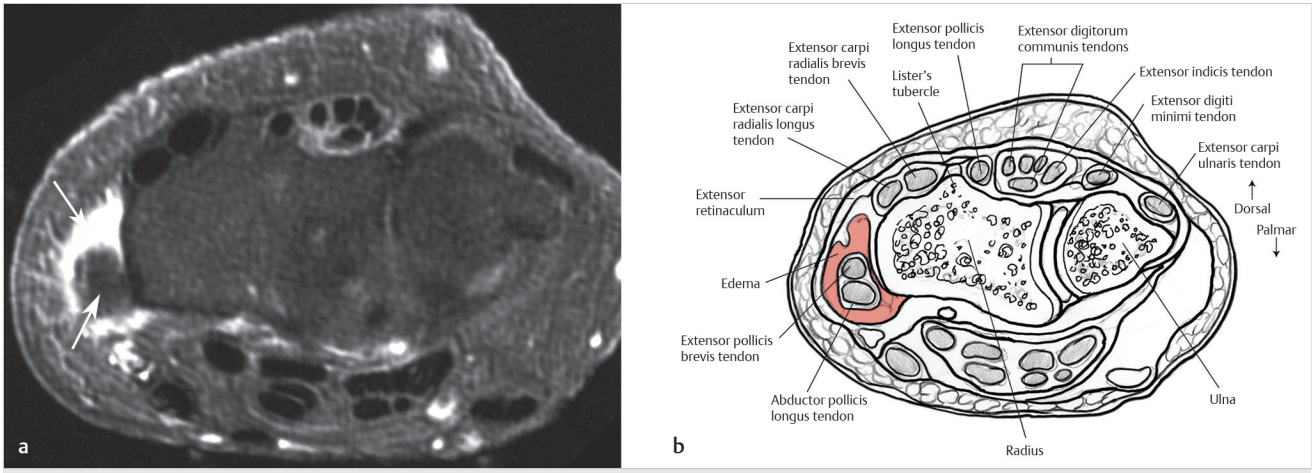


Fig. 9.11 De Quervain tenosynovitis. An axial fat-suppressed T2-weighted image **(a)** and illustration **(b)** showing thickening of the first compartment tendons and surrounding edema (*arrows on* **a**). (Reproduced with permission from Brunton LM, Anderson MW, Chhabra AB. The wrist and hand. In: Khanna AJ, ed. MRI for orthopaedic surgeons. New York: Thieme; 2010:129–143.)

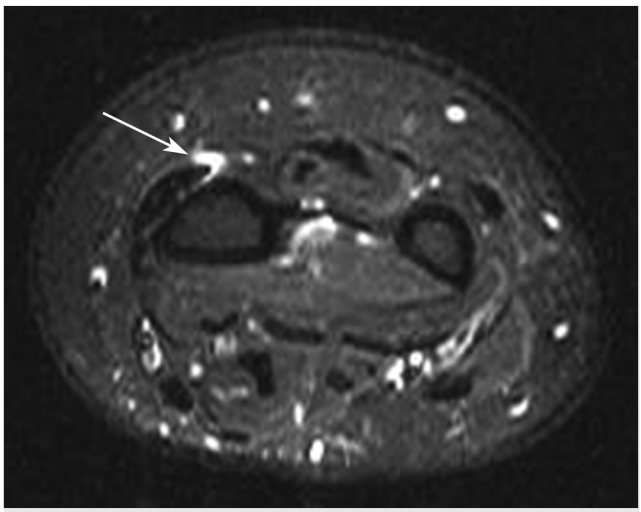


Fig. 9.12 Axial fat-suppressed T2-weighted image showing peritendinous fluid at the level where the tendons of the first and second extensor compartments cross, consistent with intersection syndrome from chronic overuse (*arrow*). (Reproduced with permission from Brunton LM, Anderson MW, Chhabra AB. The wrist and hand. In: Khanna AJ, ed. MRI for orthopaedic surgeons. New York: Thieme; 2010:129–143.)

pollicis longus, flexor carpi radialis, and flexor carpi ulnaris (▶ Fig. 9.12). Tendon subluxation or dislocation from chronic overuse may be detected on axial images. MRI may help differentiate chronic tenosynovitis and soft tissue masses such as giant cell tumor of the tendon sheath.

The clinician should keep in mind that when a tendon lies at approximately 55 degrees relative to the direction of the external magnetic field during scanning, artifactual increased intratendinous signal intensity may be observed on some imaging sequences secondary to the *magic angle phenomenon*, not to true pathology.[37,38] This phenomenon is recognized easily because the signal disappears and the tendon appears normal on heavily T2-weighted images. This phenomenon also is seen in the shoulder where the presence of the magic angle phenomenon may be mistaken for rotator cuff tendinopathy (see Chapter 7, The Shoulder).

Given that the clinical findings are typically diagnostic, MRI is not necessary for the diagnosis of stenosing tenosynovitis of the digits (trigger finger), although it may have a role in evaluating patients with postoperative recurrence or in ruling out other causes of symptoms such as tendon sheath tumor, rheumatoid synovitis, or volar plate pathology.

9.5.3 Carpal Instability

The most common pattern of carpal instability is dorsal-intercalated segment instability, often associated with disruption of the scapholunate and volar extrinsic carpal ligaments. Volar intercalated segment instability from lunotriquetral ligament deficiency is far less common. Clinically, carpal instabilities usually present as chronic wrist pain months to years after traumatic injury or secondary to inflammatory arthropathy. Physical examination findings are often equivocal, and standard lateral wrist radiographs may not always show abnormal carpal alignment. In patients with static carpal instability, the malalignment will be evident on sagittal images, and MRI may also detect associated abnormalities of the stabilizing carpal ligaments such as laxity, thickening, retraction, edema, or absence. MR arthrography and thin-section coronal gradient-echo images are the most useful for identifying the pathology of these structures. Perforations of the central portions of the interosseous ligaments may be asymptomatic and consistent with normal aging.[39]

9.6 Other Pathologic Conditions

MRI is useful in evaluating other pathologic conditions affecting the wrist and hand such as infections, osteonecrosis, RA, compression neuropathies, and soft tissue masses.

9.6.1 Infectious Conditions

Soft tissue abscesses within the hand or wrist typically show intermediate to low signal intensity on T1-weighted images and high signal intensity on T2-weighted images. A thick, enhancing wall is usually seen after the administration of intravenous contrast. MRI can evaluate the extent of soft tissue involvement and the presence or absence of osteomyelitis. An emergent MRI may be necessary in select patients with suspected necrotizing fasciitis, for which T2-weighted images may help confirm the diagnosis with increased signal intensity seen in the deep fascia, although this imaging finding is nonspecific. Given the need for urgent treatment of this condition, the clinician should also carefully evaluate for the presence of crepitus within the soft tissues and rapidly spreading erythema and induration. Prompt treatment of infectious processes in critically ill patients should never be delayed while awaiting an MRI study. MRI is of limited use in monitoring the progress of infections during the course of treatment because many of the changes observed on imaging resolve very slowly after successful treatment.[40]

9.6.2 Kienböck Disease

Idiopathic osteonecrosis of the lunate is known as Kienböck disease. This diagnosis must be considered in patients with subacute or chronic dorsal wrist pain of unknown cause. A strong association with negative ulnar variance exists, and the cause of the disorder may stem from chronic microtrauma secondary to increased shear stress on the lunate. Lichtman et al[41] defined four stages of Kienböck disease:

- Stage I: normal on conventional radiographs, changes in the lunate on MRI.
- Stage II: lunate sclerosis on conventional radiographs.
- Stage III: lunate collapse.
- Stage IV: pancarpal arthritis.

Early diagnosis by MRI is critical because nonoperative treatment and surgery are most successful in the initial stages of the disease, long before progression to lunate collapse or pancarpal arthritis.

In the first stage of Kienböck disease, MRI shows distinct changes in the lunate. On T1-weighted images, abnormally low

lunate vascularity leads to diffuse low signal intensity of the bone marrow compared with the surrounding carpal bones (▶ Fig. 9.13). This diffuse involvement distinguishes Kienböck disease from focal T1-weighted signal loss secondary to ulnar impaction syndrome, fracture, or tumor. The findings on T2-weighted images have been correlated with prognosis, showing low signal intensity with complete marrow fibrosis and increased signal intensity with lunate revascularization. When stage-I Kienböck disease is diagnosed, patients are typically immobilized for up to 3 months, and follow-up MRI examinations are used to monitor the success of nonoperative treatment. Normalization of T1 and T2 signal intensity has also been observed on MRI postoperatively in patients undergoing surgery for Kienböck disease.[42]

MRI findings are similar in idiopathic osteonecrosis of the scaphoid, also known as Preiser disease, and other carpal bones. However, because of its relative rarity, Preiser disease has been less well-characterized. One study supports the possibility of two patterns of Preiser disease based on disparate MRI patterns of vascular impairment[43]:

- Type-1 variant, diffuse involvement of the scaphoid.
- Type-2 variant, only partial necrosis.

Another study describes stages rather than patterns of Preiser disease[44]: an initial stage with proximal osteonecrosis but unaltered overall scaphoid morphology, an advanced stage with pathologic fractures and volume loss of the proximal pole, and a final stage with the entire scaphoid involved. The recognition of these distinct imaging patterns has implications for treatment, and the clinical outcome may be less favorable for patients with more diffuse involvement.[43]

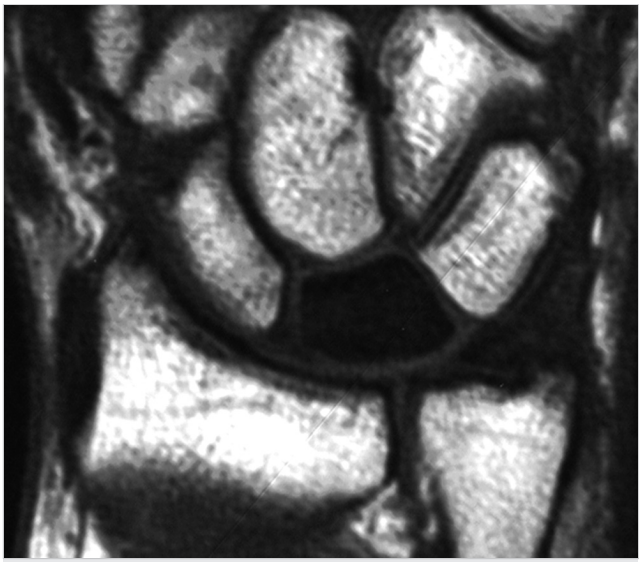


Fig. 9.13 Coronal T1-weighted image showing complete loss of normal marrow signal within the lunate, indicating necrosis (Kienböck disease). (Reproduced with permission from Brunton LM, Anderson MW, Chhabra AB. The wrist and hand. In: Khanna AJ, ed. MRI for orthopaedic surgeons. New York: Thieme; 2010:129–143.)

9.6.3 Posttraumatic Scaphoid Osteonecrosis

The tenuous retrograde blood supply of the scaphoid often results in delayed union, nonunion, and/or osteonecrosis of the proximal pole. These complications are minimized with the early diagnosis of a scaphoid fracture and the implementation of appropriate treatment. In patients with scaphoid nonunion, determination of bone viability is critical to subsequent management. Vascularity of the proximal fragment has been shown to be an important determinant of outcome after surgical reconstruction of a scaphoid nonunion.[45] Patients with proximal pole osteonecrosis may require vascularized bone grafting. Visual inspection of intraoperative punctate bleeding typically corresponds with viability of the proximal pole. Because conventional radiographs are unreliable for detecting bone vascularity, MRI can be used to assess the degree of osteonecrosis before surgery. Osteonecrosis is indicated by low signal intensity on T1-weighted images, in stark contrast to the high signal intensity of the marrow of the normal surrounding carpal bones (▶ Fig. 9.14). T2-weighted images have been shown to be unreliable for determining whether osteonecrosis is present.[46] Various authors have investigated the use of dynamic gadolinium-enhanced MRI in determining proximal pole viability and have not found it to add predictive value compared with unenhanced or standard enhanced images.[46,47]



Fig. 9.14 A coronal T1-weighted image showing loss of normal marrow signal in the proximal pole of the scaphoid (*arrow*), indicative of posttraumatic osteonecrosis. (Reproduced with permission from Brunton LM, Anderson MW, Chhabra AB. The wrist and hand. In: Khanna AJ, ed. MRI for orthopaedic surgeons. New York: Thieme; 2010:129–143.)

9.6.4 RA

RA, the most common form of inflammatory arthritis, is characterized by inflammation of the synovial lining of joints and tendon sheaths. It frequently affects the hand and wrist early in the disease process. Although RA may become a severely disabling condition, recent clinical trials unequivocally support early intervention with aggressive pharmacologic therapy for the prevention of rapid joint erosion and the improvement of long-term outcomes.[48] The necessity for timely and accurate diagnosis is implicit within this strategy. For many patients, this diagnosis is achieved only by history, physical examination, laboratory data, and conventional radiographic findings. A more expeditious diagnosis of early RA, however, may be provided with MRI.

Several reports have shown the value of MRI in detecting bone erosion and active synovitis before changes are visible on conventional radiographs.[49,50] MRI bone edema in patients with early RA has also been associated with radiographic progression.[51] Contrast-enhanced images can be obtained to investigate a possible diagnosis of RA via MRI. Intense bilateral contrast enhancement in the wrist, metacarpophalangeal joint, or proximal interphalangeal joint suggests active synovitis and indicates inflammatory arthritis (▶ Fig. 9.15). In addition, adjacent bone edema or frank erosions can be detected (▶ Fig. 9.16). These imaging findings are nonspecific; therefore, the clinician must consider the entire clinical picture so that other diagnoses (e.g., systemic lupus erythematosus, scleroderma, polymyositis, dermatomyositis, Behçet disease, or mixed connective disease) are excluded.[52,53] Advanced techniques have been developed to quantify the volume of affected synovium and bone erosion with MRI. These methods may be useful in predicting disease progression and monitoring the response to treatment.[54]

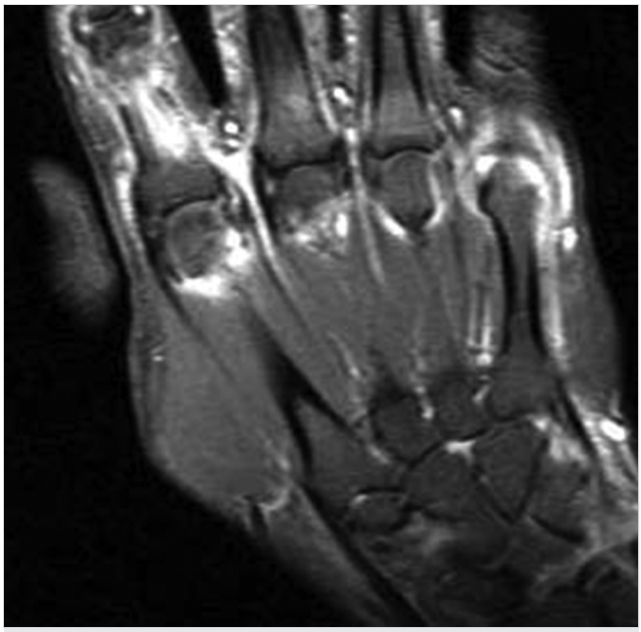


Fig. 9.15 A coronal fat-suppressed postgadolinium T1-weighted image showing diffuse high signal enhancement surrounding the metacarpophalangeal joints, consistent with synovitis from early rheumatoid arthritis (RA). Conventional radiographs were normal. (Reproduced with permission from Brunton LM, Anderson MW, Chhabra AB. The wrist and hand. In: Khanna AJ, ed. MRI for orthopaedic surgeons. New York: Thieme; 2010:129–143.)

9.6.5 Compression Neuropathies

The nerve branches in the hand and wrist most frequently affected by compression syndromes are the median nerve in the carpal tunnel and the ulnar nerve in Guyon canal. The superficial sensory branch of the radial nerve is affected much less commonly. Compression neuropathies may be triggered by intrinsic or extrinsic sources. Most patients are diagnosed by history and physical examination findings, which are then corroborated by electrodiagnostic testing. Although T2-weighted axial images may show increased intraneural signal because of nerve compression, MRI plays a primarily adjunctive role in evaluating compression neuropathies. Some patients present with atypical or equivocal clinical features or nerve conduction



Fig. 9.16 A sagittal gradient-echo image of the wrist showing multiple subchondral cysts and periarticular erosions in the setting of advanced rheumatoid arthritis (RA). (Reproduced with permission from Brunton LM, Anderson MW, Chhabra AB. The wrist and hand. In: Khanna AJ, ed. MRI for orthopaedic surgeons. New York: Thieme; 2010:129–143.)

study results, and an MRI examination may be indicated for the investigation of atypical causes such as nerve sheath tumors or extrinsic space-occupying lesions. Another indication for MRI is recurrent symptoms after surgery in patients with incomplete decompression, excessive scarring, hematoma formation, or another space-occupying lesion, for whom additional treatment can be initiated accordingly (▶ Fig. 9.17).

In the evaluation of the median nerve at the carpal tunnel, T1-weighted and T2-weighted axial images are obtained to screen for obvious anatomic anomalies or pathologic findings. The median nerve is examined for changes in size, shape, or signal intensity. A predominant finding in patients with carpal tunnel syndrome is increased signal within the median nerve on T2-weighted images, which is related to chronic compression or ischemia. Other detectable abnormalities include bowing of the flexor retinaculum, deep palmar bursitis, tenosynovitis, or the presence of a space-occupying soft tissue mass. Intraneural lesions usually are well delineated. Although dynamic and motion studies of the median nerve in the carpal tunnel have been described, they are rarely used.[55]

In the presence of ulnar nerve compression at the wrist, MRI may elucidate the cause when other diagnostic methods do not. Several causes include posttraumatic neuritis, occult fractures, pisotriquetral arthropathy, anatomic variants or ganglion cysts, and other soft tissue masses (▶ Fig. 9.18). Again, axial images are best for evaluating the ulnar nerve in Guyon canal. The information gained from MRI may help guide treatment decisions, with surgical decompression reserved for patients in whom an extrinsic source of compression is found.

In chronic cases of any nerve compression syndrome, MRI may show patterns of muscle atrophy consistent with denervation of the affected nerve. Denervated muscles typically show increased signal intensity on T1-weighted images, representing fatty replacement in the more chronic setting, or on fat-suppressed T2-weighted sequences, representing muscle edema in the more acute to subacute phases.

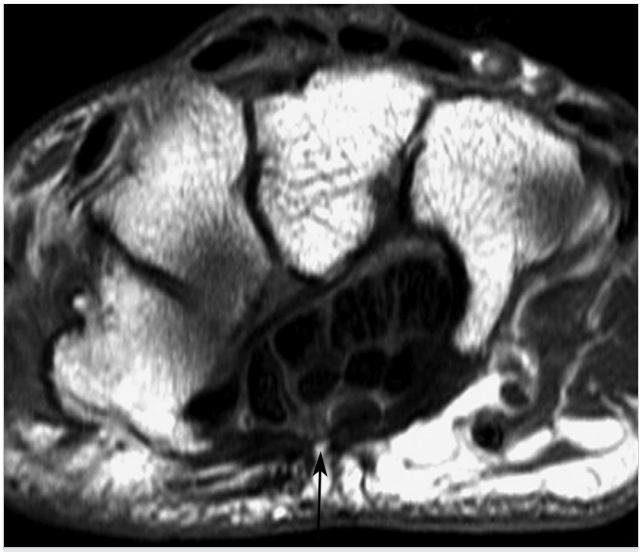


Fig. 9.17 A postoperative axial T1-weighted image at the level of the carpal tunnel showing incomplete release of the transverse carpal ligament (*arrow*). (Reproduced with permission from Brunton LM, Anderson MW, Chhabra AB. The wrist and hand. In: Khanna AJ, ed. MRI for orthopaedic surgeons. New York: Thieme; 2010:129–143.)

9.6.6 Soft Tissue Masses

Soft tissue masses occur commonly in the wrist and hand. Many of these lesions such as ganglia can be managed reliably without the use of advanced imaging. Nevertheless, MRI remains the standard for evaluating soft tissue masses of unclear clinical diagnoses. The primary role of MRI is in differentiating determinate from indeterminate lesions (see Chapter 19, Soft Tissue Tumors, and Chapter 20, Bone Tumors).[56] Determinate lesions may be diagnosed on MRI by a radiologist or clinician with a high degree of certainty on the basis of the imaging characteristics. Indeterminate lesions are those for which the radiologist or clinician is unable to provide a definitive diagnosis and which therefore warrant biopsy. Intravenous contrast may help in distinguishing solid and cystic lesions.

Benign solid lesions of the wrist and hand with characteristic MRI features include hemangioma, giant cell tumor of the tendon sheath, lipoma, Dupuytren disease, hamartoma, peripheral

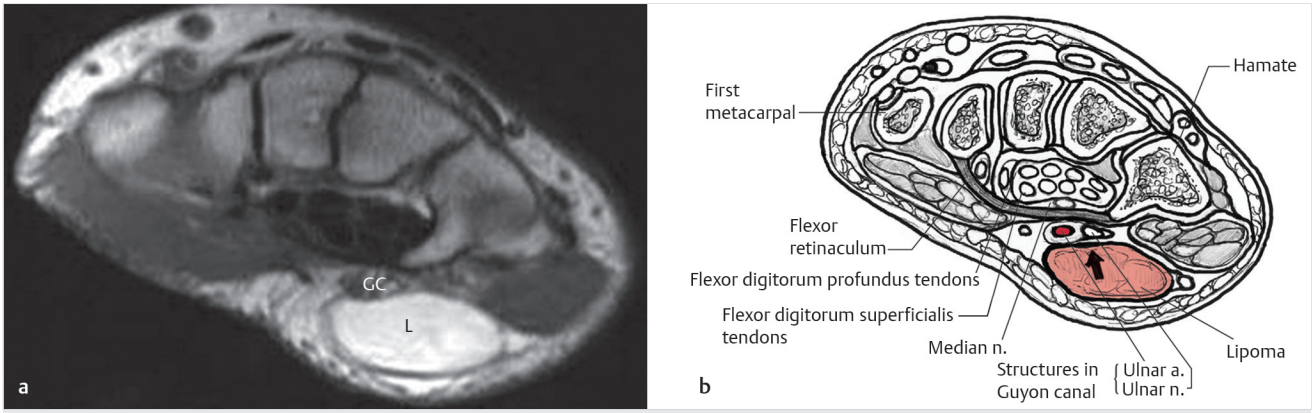


Fig. 9.18 Ulnar tunnel syndrome. An axial T1-weighted image **(a)** and illustration **(b)** showing a large well-circumscribed lipoma (*L*) compressing Guyon canal (GC). (Reproduced with permission from Brunton LM, Anderson MW, Chhabra AB. The wrist and hand. In: Khanna AJ, ed. MRI for orthopaedic surgeons. New York: Thieme; 2010:129–143.)

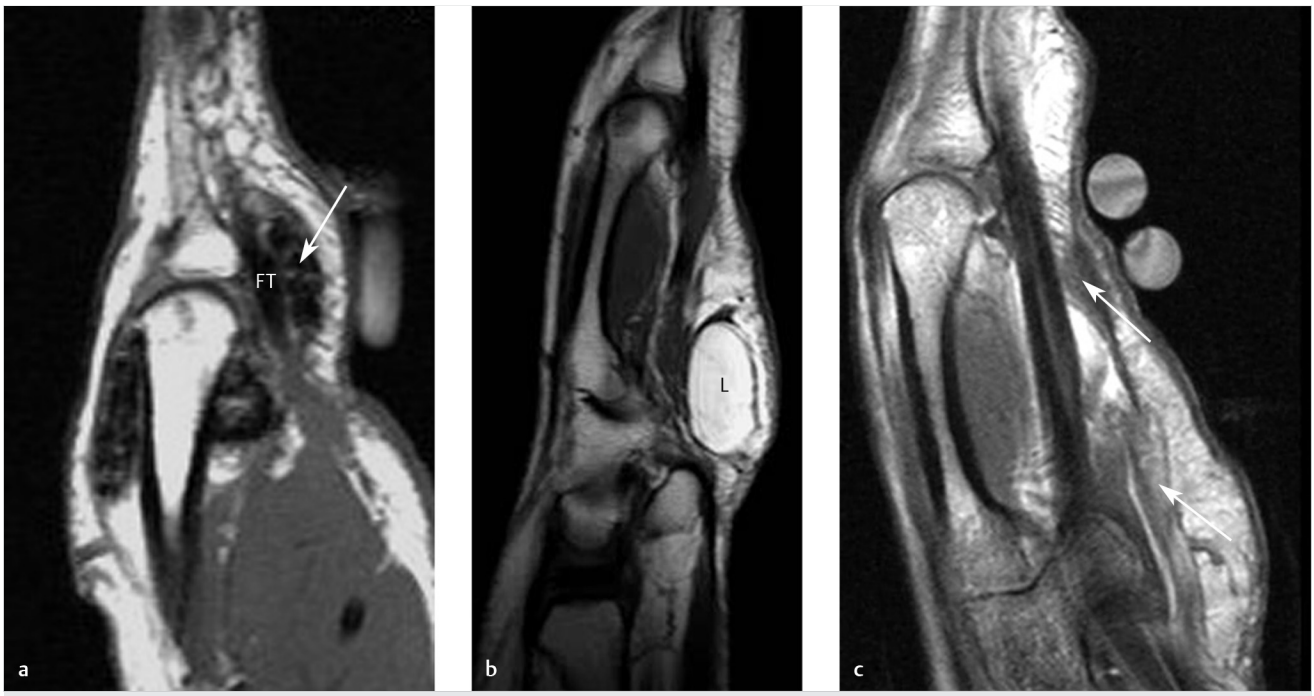


Fig. 9.19 Benign, soft tissue tumors of the hand. **(a)** Giant cell tumor of the tendon sheath. A sagittal T1-weighted image of the hand and digit depicting a solid mass (*arrow*) of mixed signal intensity originating from the underlying flexor tendon (*FT*). **(b)** Lipoma. A sagittal T1-weighted image showing a well-circumscribed lipoma (*L*) in the distal palm. Note how the signal characteristics of the lipoma match those of surrounding subcutaneous fat. **(c)** Dupuytren contracture. A sagittal T1-weighted image showing thickened palmar fascial bands (*bottom arrow*) and a fibrotic nodule just proximal to the metacarpophalangeal joint (*top arrow*) in a patient with Dupuytren disease. The circular structures are fiduciary markers that delineate the site of the patient's symptoms. (Reproduced with permission from Brunton LM, Anderson MW, Chhabra AB. The wrist and hand. In: Khanna AJ, ed. MRI for orthopaedic surgeons. New York: Thieme; 2010:129–143.)

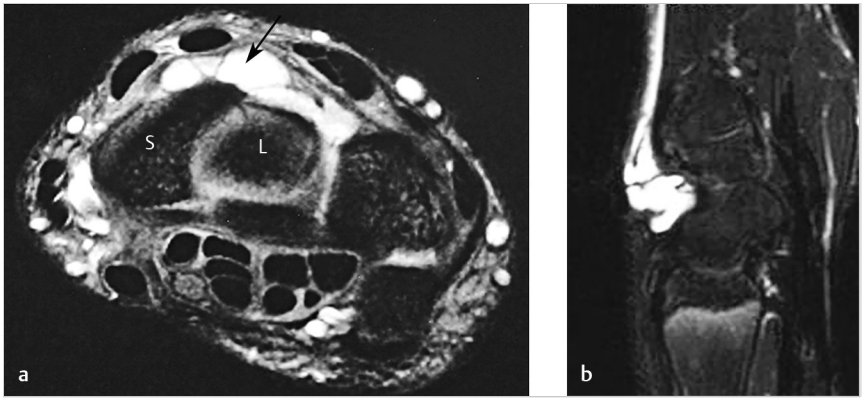


Fig. 9.20 Ganglion cyst. **(a)** An axial short tau inversion recovery (STIR) image of the wrist showing a multiloculated ganglion cyst (*arrow*) adjacent to the dorsal margin of the scapholunate articulation. *S*, scaphoid; *L*, lunate. **(b)** A sagittal STIR image depicting a dorsal ganglion cyst. (Reproduced with permission from Brunton LM, Anderson MW, Chhabra AB. The wrist and hand. In: Khanna AJ, ed. MRI for orthopaedic surgeons. New York: Thieme; 2010:129–143.)

nerve sheath tumor, and glomus tumor (▶ Fig. 9.19). Although malignant, soft tissue masses are rare in the hand and wrist, MRI may show irregular features or tissue necrosis that suggests a malignant process and requires additional evaluation before excision.[57] Many malignant lesions, however, have a nonaggressive appearance on MR images. The value of MRI in the assessment of bone malignancies is to identify and characterize the extent of soft tissue involvement.

Ganglion cysts usually can be diagnosed clinically with palpation and visualization. However, occult ganglions can cause chronic dorsal wrist pain or nerve compression. Conventional radiography is usually normal. MRI may help in differentiating ganglion cysts from synovitis in the chronically painful wrist. There are important differences between the two entities with regard to margins, shape, specific site of involvement, dimensions, internal structure, and contrast enhancement.[58] Ganglia have defined margins, a spherical shape, a septated, multilocular appearance, and wall enhancement (▶ Fig. 9.20), whereas synovitis shows diffuse margins, a crescentic shape, lack of septa, and diffuse enhancement.[58] Distinguishing these two diagnoses in the setting of chronic wrist pain has obvious implications for treatment because patients with findings of

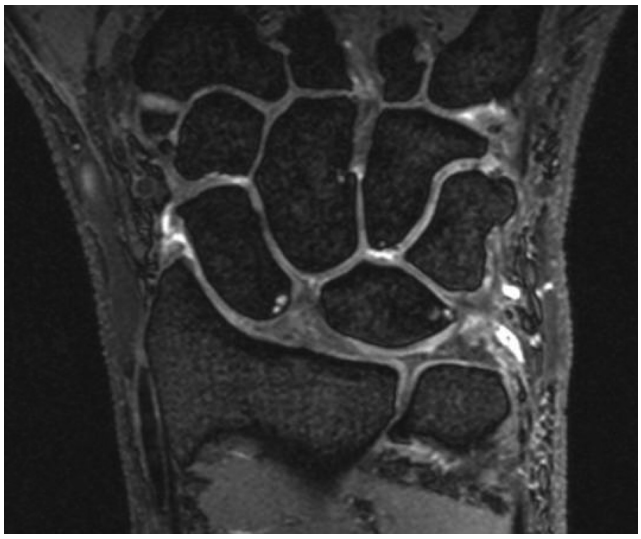
chronic synovitis typically undergo a more prolonged course of nonoperative treatment than do patients with ganglia. Approximately 30% of patients with radial-side ganglia may have associated pathology such as interosseous ligament tears that are detectable by MRI.[59]

9.7 Postoperative MRI Findings

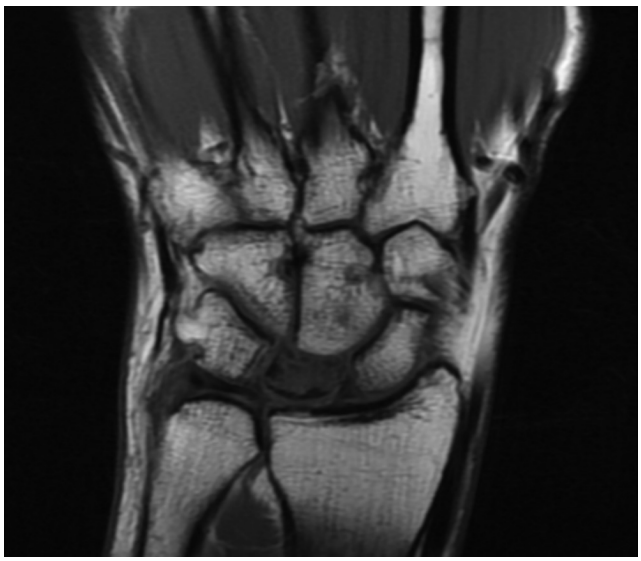
Interpretation of postoperative MRI of the hand and wrist can be challenging because of the presence of postsurgical changes and, in some cases, orthopaedic hardware. If metal implants are present, special consideration is needed to reduce susceptibility artifacts, which cause signal void, pile-up, distortion, and loss of fat suppression. When using conventional MR sequences, FSE and STIR sequences produced with a high bandwidth and thin sections tend to be most resistant to metal artifact, whereas gradient-echo sequences should be avoided.[60] Newer multispectral MRI techniques specifically designed to reduce metal artifact, such as multiacquisition variable-resonance image combination (MAVRIC) and slice encoding for metal artifact correction (SEMAC), are highly effective.[61] Nonetheless, MRI has an established role in evaluating patients after hand and wrist surgery. After scaphoid fixation or vascularized bone grafting, MRI with or without gadolinium enhancement can be used to evaluate proximal pole viability.[62,63] In patients with recurrent or persistent symptoms after triangular fibrocartilage complex or intrinsic ligament repair, MRI can be used to evaluate the integrity of the repair and may show a persistent defect or a new tear.[64] After tendon repair or transfer, MRI can be used to detect tendon ruptures, peritendinous adhesions, and pulley injuries in patients with continued motion loss.[65] MRI can also be useful in patients with compressive neuropathy who have not improved after nerve release. After carpal tunnel release, MRI findings should include a complete defect in the flexor retinaculum and transverse carpal ligament with enlargement of the carpal tunnel and volar displacement of the flexor tendons and median nerve.[66] In patients with continued or recurrent symptoms, MRI may show incomplete release of the transverse carpal tunnel, a ganglion cyst or other structural lesion compressing the nerve, or structural or signal abnormalities within the nerve itself.

9.8 Common Clinical Questions

1. A 58-year-old man presents with ulnar-sided wrist pain that is especially bothersome during his golf swing. A T2-weighted MR image is shown below. What is the most likely cause of his discomfort?
   a) Scapholunate ligament injury.
   b) Triangular fibrocartilage complex injury.
   c) Pisotriquetral arthritis.
   d) Distal radioulnar arthritis.



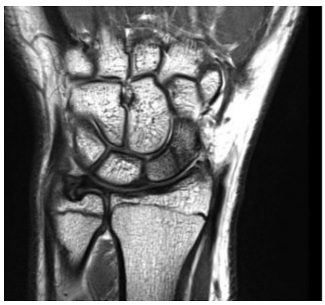
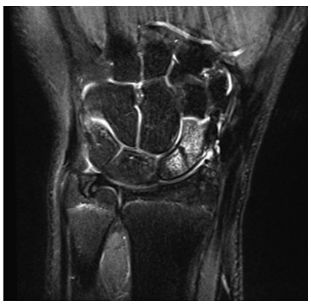
2. A 41-year-old woman presents for evaluation of a 10-year history of wrist pain. The pain has become worse over time. She denies any history of wrist injury. A T1-weighted MR image is shown. What is the most likely diagnosis?
   a) Preiser disease.
   b) Ulnar impaction syndrome.
   c) Kienböck disease.
   d) Scapholunate ligament avulsion.



3. What imaging features characterize soft tissue abscesses of the hand?
   a) High signal intensity on T1-weighed images and low signal intensity on T2-weighted images.
   b) Low signal intensity on T1-weighted and T2-weighted images.
   c) High signal intensity on T1-weighted and T2-weighted images.
   d) Low signal intensity on T1-weighed images and high signal intensity on T2-weighted images.

4. In which of the following patients is an MRI most likely to be useful in making a diagnosis?
   a) A patient with numbness in the thumb, index finger, and middle finger and evidence of carpal tunnel syndrome on electrodiagnostic testing.
   b) A patient with numbness in the small and ring fingers and evidence of cubital tunnel syndrome on electrodiagnostic testing.
   c) A patient with a 0.5 × 0.5-cm dorsal wrist mass that is nonadherent to the overlying skin and transilluminates.
   d) A patient with numbness in the thumb, index finger, and middle finger that has recurred after carpal tunnel release with evidence of carpal tunnel syndrome on electrodiagnostic testing.

5. A 17-year-old boy has had persistent wrist pain since falling on an outstretched hand while skateboarding 7 months ago. This is the first time he has sought treatment. T1-weighted and T2-weighted precontrast MR images are shown below. Postcontrast images show uniform enhancement of the scaphoid. What is the diagnosis?
   a) Posttraumatic scaphoid osteonecrosis.
   b) Scaphoid fracture nonunion.
   c) Healed scaphoid fracture.
   d) Preiser disease.

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Answers to Common Clinical Questions

1. Answer: b. The MRI shows a central perforation of the triangular fibrocartilage complex (TFCC). Although TFCC abnormalities can be found in asymptomatic individuals, this patient's history of ulnar-sided pain exacerbated by repetitive wrist pronation and supination (as occurs in golfing) is highly consistent with the diagnosis of a TFCC tear.
2. Answer: c. The MRI shows diffusely low signal intensity of the lunate compared with the surrounding carpal bones on T1-weighted imaging, consistent with the diagnosis of Kienböck disease.
3. Answer: d. Soft tissue abscesses within the hand typically are characterized by low signal intensity on T1-weighted images and high signal intensity on T2-weighted images.
4. Answer: d. Most patients with carpal tunnel and cubital tunnel syndrome are diagnosed on the basis of history and physical examination findings, which are corroborated by electrodiagnostic testing. Many ganglion cysts can also be diagnosed clinically. However, MRI is indicated in patients who have recurrent symptoms after undergoing carpal or cubital tunnel release to evaluate for incomplete decompression, excessive scarring, hematoma formation, or another space-occupying lesions.
5. Answer: b. This is a scaphoid fracture nonunion. Low signal intensity on T1-weighted and T2-weighted images would be suggestive of osteonecrosis. In this case, enhancement with the addition of contrast confirms scaphoid vascularity.